SMART INDIA HACKATHON 2026

InfraSense: Indigenous Atmospheric Infrasound Monitoring Platform

- **Problem Statement ID –** SIH26144
- **Problem Statement Title -** High Sensitivity Micro-Barometer Infrasound sensor
- **Theme-** Smart Automation
- **PS Category-** Hardware
- **Team ID-** 155882
- **Team Name-** SPADES

SPADES

PROBLEM STATEMENT & PROPOSED SOLUTION

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PROBLEM
Weak infrasound signals are buried in wind and weather noise, and imported micro-barometers are costly.

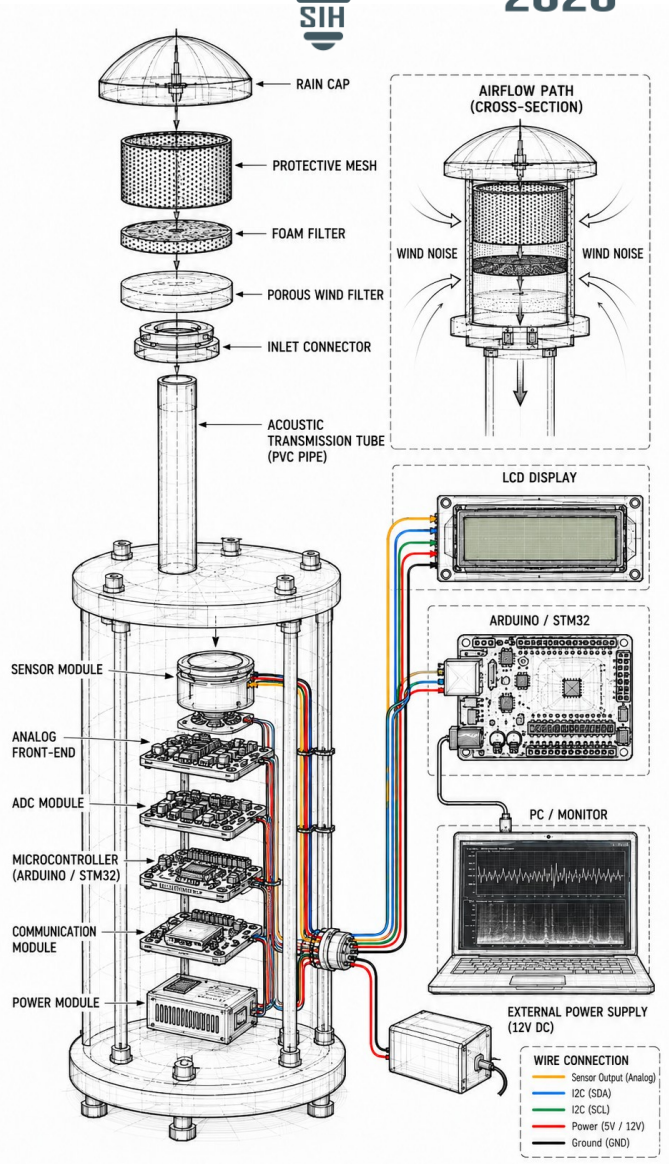
OUR IDEA
A low-cost indigenous micro-barometer that filters wind noise and detects events live on a dashboard.

CHALLENGES

- **Wind Turbulence**
- **Temperature Drift**
- **Weather Pressure Variations**
- **Weak Infrasound Signals**
- **Analog Signal Integrity**
- **Long-Term Pressure Stability**
- **Event Identification**
- **Calibration Requirement**
- **Sensor Maintenance**

PROPOSED SOLUTIONS

- Multi-stage Passive Wind Suppression
- Underground Environmental Enclosure + Temperature Compensation (Planned)
- Differential Pressure Equalization Network
- MEMS Capacitive Sensor + Low Noise Amplifier
- Low-Noise Analog Front-End + 24-bit ADC Digitizer
- Long-Period Pressure Equalization Network
- FFT-Based Signal Analysis
- Reference Calibration using Pressure Testing
- Modular Cartridge Architecture



KEY INNOVATION

- ☑ **Differential Measurement**
- ☑ **Pressure Equalization**
- ☑ **Reference Calibration**
- ☑ **STA/LTA Detection**

TECHNICAL APPROACH

SYSTEM WORKFLOW

Pneumatic front end | Electronics | Software

01 Atmospheric Inlet → 02 Wind Noise Reduction Array → 03 Acoustic Tube → 04 Underground Enclosure → 05 Differential Pressure Chamber → 06 Long-Period Pressure Equalization Network → 07 MEMS Capacitive Pressure Sensor

07 MEMS Capacitive Pressure Sensor → 08 Low Noise Amplifier → 09 24-bit ADC Digitizer → 10 STM32 Controller → 11 Filtering & FFT Analysis → 12 STA/LTA Event Detection → 13 Event Storage & Alerts → 14 Real-Time Dashboard

HARDWARE

- MEMS Capacitive Pressure Sensor
- MEMS Diaphragm-Based Mechanical Transducer
- Differential Pressure Chamber
- Pressure Equalization Network
- Wind Noise Reduction Module
- STM32 + 24-bit ADC Digitizer
- WiFi Telemetry

SOFTWARE

- Embedded C on STM32
- Butterworth IIR filtering
- Temperature Compensation
- Radix-2 FFT, Welch PSD
- STA/LTA event detection
- MQTT telemetry
- Next.js + React (TypeScript)
- HTML5 Canvas live charts

TARGET EVENT TYPES

- Explosion
- Rocket Launch
- Thunderstorm
- Ambient Noise

3-LAYER APPROACH

1 Pneumatic Front End ▶ 2 Electronics ▶ 3 Software & Dashboard

GitHub: Repository **Live:** Dashboard
Progress: 60% – sensor + live dashboard working

YouTube Demo

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

- ✓ Commercial Components Available
- ✓ Standard PCB Manufacturing
- ✓ Modular Design
- ✓ Real-Time Monitoring
- ✓ Scalable Architecture

Feasibility criteria met **5 / 5**

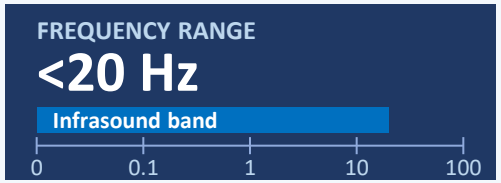
CHALLENGE VS SOLUTION

Challenge		Engineering Solution
Pressure Drift	→	Differential Pressure Equalization
Thermal Drift	→	Underground Environmental Enclosure
Long-Term Stability	→	Self-Calibration using Reference Pressure Testing
Maintenance	→	Plug-and-Replace Cartridge Design

VALIDATION & EVALUATION PLAN

- ✓ Low-Frequency Signal Detection
- ✓ Frequency Response Testing
- ✓ Noise Floor Measurement
- ✓ Sensitivity Estimation
- ✓ Stability Testing

TARGET PERFORMANCE



- Low Drift
- Real-Time Monitoring
- 0.01–20 Hz Response
- Modular Maintenance

COMMERCIAL FEASIBILITY

UNIT COST
₹30K–60K per prototype
₹0.8–1.5 L per field station (est.)

TARGET BUYERS
IMD, NCS, DRDO, ISRO, NTRO, research labs

SCALING PATH
Standard PCB + modular cartridge allow low-cost batch builds

TIMELINE
Prototype: 6 months
Field pilot: 12–18 months (est.)

IMPACT AND BENEFITS

DIRECT IMPACT ON TARGET USERS

Weather & Disaster Agencies
- Atmospheric anomaly detection
- High-energy event monitoring

Geoscience Researchers
- Monitor atmospheric signatures of geophysical events
- Support scientific research

Space Launch Centres
- Detect launch-generated infrasound
- Support aerospace testing

Climate & Volcano Scientists
- Monitor volcanic activity
- Study atmospheric dynamics

Industrial Plant Operators
- Monitor large industrial events
- Detect abnormal pressure disturbances

ATMOSPHERIC SIGNALS → **Indigenous Infrasound Monitoring Platform** → **ACTIONABLE MONITORING INFORMATION**

STRATEGIC IMPACT & ECONOMIC BENEFITS

Sustainable Development

SDG 9 — **Industry, Innovation & Infrastructure**

SDG 16 — **Peace, Justice & Strong Institutions**

SUPPORTS
- Atmanirbhar Bharat
- Indigenous Strategic Technology Development

Technical Benefits
- High-Sensitivity Infrasound Detection
- Low-Noise Signal Conditioning & Digitization
- Differential Pressure Measurement
- Real-Time Monitoring & Analysis

Economic Benefits
- Modular Maintenance Architecture
- Reduced Lifecycle Servicing Cost
- Commercially Available Components
- Scalable Sensor Deployment

Environmental Benefits
- Low-Power Operation
- Long-Term Field Deployment
- Minimal Environmental Footprint
- Supports Atmospheric Research

Strategic Benefits
- Atmospheric Event Intelligence
- Strategic Monitoring Capability
- Supports Strategic Monitoring Applications
- Indigenous Monitoring Infrastructure

RESEARCH AND ANALYSIS

LITERATURE SURVEY (RESEARCH-DRIVEN DESIGN)

CTBTO Infrasound Monitoring Architecture
Reference IMS station design for sensor and array layout
Source: CTBTO Website

Wind Noise Reduction Systems
Porous-hose and pipe-array filters to suppress wind noise
Source: Research Paper

Differential Pressure Measurement
Inlet vs. sealed reference volume, sensor reads the difference
Source: Academic Research

Pressure Equalization Networks
Slow leak path that removes daily barometric changes
Source: Academic Research

Low-Noise Atmospheric Sensing
Low-noise front end and signal-to-noise design
Source: Published Literature

Strategic Monitoring Applications
Explosion, launch and volcano monitoring use cases
Source: CTBTO / Research Literature

FIELD TESTS & SIMULATION RESULTS



CURRENT STATUS
- ✓ **Dashboard:** simulated data demo
- ✓ **Processing:** implemented
- ○ **Hardware tests:** planned
- ○ **Noise floor, sensitivity:** to be measured

Benchmark target: <3 mPa (Chaparral M25)

- ✓ Live Waveform Display
- ✓ FFT Spectrum Analysis
- ✓ Frequency-Time Spectrogram
- ✓ STA/LTA Event Detection
- ✓ Event Logging & Alerts
- ✓ Station Health Monitoring

GAP & PROBLEM IDENTIFICATION
Need indigenous high-sensitivity sensors
Source: IMD Pune indigenous system (Hindustan Times)

ECONOMIC & STRATEGIC LANDSCAPE
Demand: weather, defence, space, research
Source: IMD Surface Instruments Division

COMPETITIVE ANALYSIS & BENCHMARKING
vs Chaparral M25 (imported): indigenous, modular
Source: Chaparral Physics Model 25 datasheet

POLICY & ECOSYSTEM ANALYSIS
Atmanirbhar Bharat · Make in India
Source: IMD indigenous observation system (HT)

Developed Monitoring Portal: Micro-Barometer Dashboard